

5	(a) (i)	resistance = V/I	B1	
		very high/infinite resistance at low voltages	B1	
		resistance decreases as V increases	B1	[3]
	(ii)	p.d. from graph 0.50 (V)	C1	
		resistance = $0.5/(4.4 \times 10^{-3})$		
		= 110 (114) Ω	A1	[2]
	(b) (i)	current (= $1.2/375$) = 3.2×10^{-3} A	A1	[1]
	(ii)	current in diode = 4.4×10^{-3} (A)		
		total resistance = $1.2/4.4 \times 10^{-3} = 272.7$ (Ω)	C1	
		resistance of $R_1 = 272.7 - 113.6 = 160$ (159) Ω	A1	
		or		
		p.d. across diode = 0.5 V and p.d. across $R_1 = 0.7$ V	(C1)	
		resistance of $R_1 = 0.7/4.4 \times 10^{-3}$		
		= 160 (159) Ω	(A1)	[2]
	(iii)	power = IV or I^2R or V^2/R	C1	
		ratio = $(4.4 \times 0.5)/(3.2 \times 1.2)$		
		or $[(4.4)^2 \times 114]/[(3.2)^2 \times 375]$		
		or $[(0.5)^2 \times 375]/[114 \times (1.2)^2]$		
		= 0.57	A1	[2]